

you might want to subtract the total of the numbers in cells A1 to A6 by a specific number, say, 50, which is not in any cell. Simply edit the formula and stick in your constant like in the following:

```
=SUM(A1:A6)-50
```



Multiplication And Division

These functions are quite simple to use at the outset, but there was a reason we didn't stick it in with the addition and subtraction functions. We'll look at the reason soon. First the regular functions:

```
=A1*A2/A3*D6
```

Simple enough: use the required signs when necessary. However, the answer can differ depending how the calculation is done. You can choose how it's done with the power of the brackets:

```
=(A1*A2/A3)*D6
```

Or

```
=(A1*A2)/(A3*D6)
```

Also similar to our SUM function, we have a function for multiplication called "PRODUCT". You can use it like so:

```
=PRODUCT(A1:A6)
```

You should have figured out that this function will give you the product of all cells from A1 to A6. However, there's a chance that you might encounter a problem should one of the cells you included in the above formula contain a value of 0 (which you don't want to include in the calculation), in which case your result will end up in a straight zero. You can use this solution to avoid the problem:

```
=PRODUCT(IF(A1:A3,A1:A3))
```

Careful when entering the brackets here; this formula should give you the product of the cells A1 to A3 while ignoring the zeroes.



Calculating Percentage

Again, one of the most basic, yet useful functions: you can calculate what percentage an amount is out of the total, assuming you have both values. For example, you can see scores in percentages instead of the actual scores. The formula:

```
=E4/E5
```

As usual, enter it in the cell you want the result to come into. E4 pertains to the amount, out of the total, which is E5. While this looks like a simple division formula, you can change the view to percentage by clicking on the Percentage icon on the formatting toolbar.



Increase or Decrease a Number By A Specified Percentage

There are more interesting things you can do with percentages, like increasing or decreasing your values by a set amount of percentage. For example, you can use this function if you want to

increase someone's salary by 35%:

```
=E4*(1+35%)
```

When increasing the percentage by a value situated in another cell, the "%" isn't needed as long as the cell itself contains the "%" sign. You can simply use

```
=E4*(1+E3)
```

Here you need to make sure that E3 contains a "%" after the value, like "30%". Or you can simply change the formula itself to look like

```
=E4*(1+E3%)
```

For subtracting by a certain percentage, simply replace the "+" operator with a "-" sign.



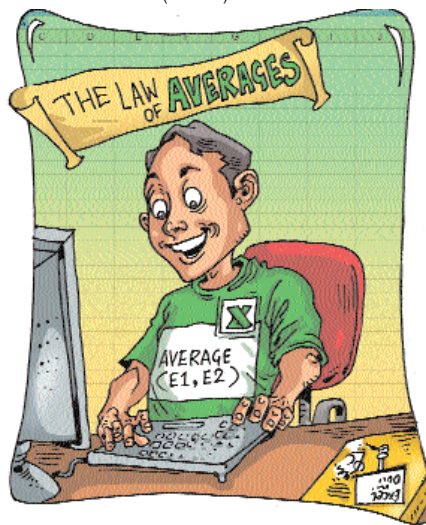
Calculating Averages

A slightly more complicated formula but still useful and easy to understand. The "AVERAGE" formula will display the average of the numbers located in several cells. This comes in most handy when calculating averages for students' mark-sheets.

```
=AVERAGE(E1,E2)
```

If your data is in a sequential line of rows or columns, you can simply enter something like:

```
=AVERAGE(E1:E3)
```



This will include all the values in the cells E1 to E3 to calculate the average. Remember, you can do this for just about any formula as long as the data is aligned sequentially. Again, similar to multiplication, you can edit this formula accordingly so that it doesn't include the zeroes or blanks that may be located in any of the cells E1 to E3:

```
=AVERAGE(IF(E1:E3,E1:E3))
```

Note: In such formulas, including the ones that follow, you might have to press {Ctrl} + {Shift} + [Enter] after typing the formula into the cell.



Interests

Interest is a function used regularly. This formula will allow you to

know the interest you have to pay every month when you have access to the info of the total loan received, the interest rate, and the number of payments you have to make. You can do this with the following formula:

```
=PMT(E2/12,E3,-E1)
```

E2 is the rate of interest, 12 indicates a monthly payment, E3 denotes the number of payments in months, and -E1 is the total amount of loan received.



Root And Square Root Functions

These functions can be useful occasionally. The syntax for this function / formula is rather simple. For square root: =E1^(1/2), and for cube root,

```
=E1^(1/3)
```

The above formulas will give you the square and cube root results of cell E1. However there's also another way to write these formulas, by directly using the square and cube functions, which is, for square root, =SQRT(E1), and for cube root, =POWER(E1;1/3).



Amount of Time Passed Since a Specified Date

On we move to the date and time formulas. This one will display, in a cell, the amount of time since a date specified in another cell. An example:

```
=YEAR(TODAY())-YEAR(E9)
```

Be careful when entering the brackets. The date entered in cells is in MM / DD / YYYY format (Month, Date, and Year) by default in Excel 2007, so make sure you get the format right. (Or else mess around with the International settings in the Control Panel.)

If you need to display the number of months or days instead of years, use the following examples. For displaying the amount of time passed in months:

```
=DATEDIF(E9,TODAY(),"m")
```

In days:

```
=DATEDIF(E9,TODAY(),"d")
```



Calculating the Number of Days in a Specified Month

Suppose you have the months listed in column A, while the year is specified in column B, and you want to find out the number of days a month contains. This formula helps you find exactly that. Move to the cell you want the result to come in and enter:

```
=DAY(DATE(B5,A5+1,0))
```

Edit your formula accordingly. In the above example, B5 is the cell containing the year, such as "2007", while A5 contains the month, such as "02". remember that Excel works with numbers, and the formula will not work if you entered "February" in cell A5 instead of "02". ■

(We continue this series next month.)